

PROJECT REPORT

Auto-Ticket Classification using Flow Designer

ServiceNow School IT Helpdesk Automation

Team Member	Primary Responsibility
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1. EXECUTIVE SUMMARY

In school environments, students and faculty regularly face routine IT difficulties such as Wi-Fi disconnection, classroom projector failures, forgotten credentials, or sluggish laboratory computers. In traditional setups, reporting these issues requires users to navigate confusing technical taxonomies or depends on manual triage by busy technicians. This creates resolution delays, incorrect routing, and unnecessary administrative burdens.

The **Auto-Ticket Classification using Flow Designer** project delivers an automated, no-code/low-code solution built entirely on the ServiceNow cloud platform. By separating the user-facing request interface from internal categorization logic, requesters simply supply their identity and a brief issue description in everyday language. Upon ticket submission, a native Flow Designer workflow triggers immediately, parses the text against configured keyword patterns, assigns the appropriate dependent Category and Subcategory, and dispatches an automated email acknowledgment to the caller.

This report documents the end-to-end design, implementation, testing, and deployment of the solution across five structured project sprints.

2. IDEATION PHASE

2.1 Empathize and Discover

The team analyzed typical user journeys and identified distinct friction points. Students experience anxiety and hesitation when faced with technical dropdowns, while helpdesk technicians spend significant portions of their shifts reclassifying misrouted tickets. The empathy mapping exercise highlighted the need for an intuitive, minimal-input interface paired with reliable background classification.

2.2 Problem Definition

Students need an effortless mechanism to report IT disruptions without having to understand technical categorization, while technicians require pre-classified, structured tickets ready for direct resolution.

2.3 Brainstorming and Idea Prioritization

The team evaluated sixteen candidate concepts across classification techniques, user experience designs, notification strategies, and platform capabilities. Each idea was scored for organizational Importance and technical Feasibility:

- **Selected Core Architecture:** Custom School IT Ticket table, Flow Designer keyword rules, dependent choices, email notification, and Update Set tracking.
- **Deferred Features:** Complex predictive AI models and external API integrations were deferred to future iterations to ensure 100% deterministic accuracy and zero external licensing dependencies.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Mapping

The user experience was mapped across five stages: Entice (issue encountered), Enter (form accessed), Engage (ticket submitted), Exit (immediate confirmation received), and Extend (helpdesk triage and resolution).

3.2 Functional Requirements (FR)

- **FR-1:** Custom table supporting ticket number, caller, description, category, subcategory, state, and assignment.
- **FR-2:** Simplified submission form requiring only Caller and Short Description.
- **FR-3:** Automated keyword evaluation for Network/Wi-Fi, Hardware/Projector, Access/Password, and Performance/Slow issues.
- **FR-4:** Automated email acknowledgment containing the ticket reference number.
- **FR-5:** Centralized helpdesk list view for technician triage.
- **FR-6:** Role-based access control (ACLs) and Update Set tracking.

3.3 Non-Functional Requirements (NFR)

The solution enforces high Usability (streamlined form), robust Security (platform ACLs), High Maintainability (flow rules updated without coding), and Deterministic Reliability.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the identified problem by shifting supported classification work from the requester to the ServiceNow workflow. The student supplies basic information that is natural to provide, while the workflow applies the internal classification rules.

For the implemented prototype, Wi-Fi or Network language maps to Network / Wi-Fi, projector-related language maps to Hardware / Projector, password or login language maps to Access / Forgot Password, and slow or hanging computer language maps to Performance / Slow Computer.

4.2 Proposed Solution

- Create the School IT Ticket custom table and School IT Support application/module.
- Capture Caller and Short Description from the requester.
- Use dependent Category and Subcategory choices.
- Trigger the classification flow when a new ticket is created with an empty Category.
- Apply branches for network, hardware, access and performance keywords.
- Send an acknowledgment email to the Caller.
- Use an Update Set to track and export the configuration for deployment.

4.3 Solution Architecture

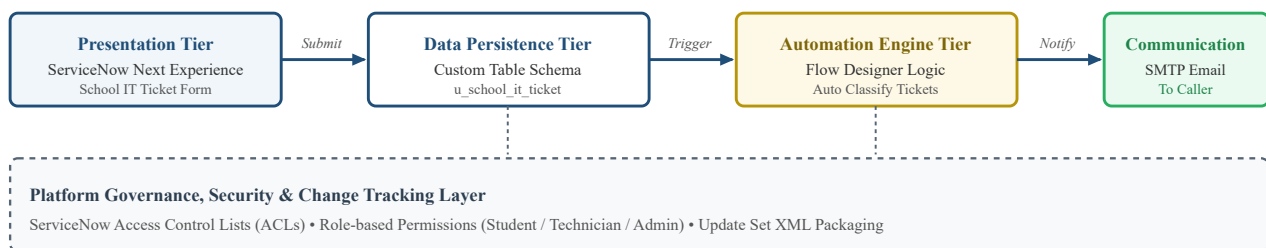


Figure 4: Solution architecture connecting the ServiceNow user interface, ticket data, Flow Designer and notification.

Solution Architecture (Figure 4) Detailed Explanation:

Figure 4 connects user interactions directly to underlying platform services. Student inputs from the UI populate the custom schema. The Flow Designer execution engine asynchronously processes the record without degrading UI responsiveness, evaluates condition logic, performs field updates, and invokes the notification subsystem.

5. DATA FLOW & PROCESS LOGIC

5.1 Data Flow Architecture

Data moves from the student requester into the School IT Ticket table, triggers the Flow Designer engine, and produces automated classification and notification records.

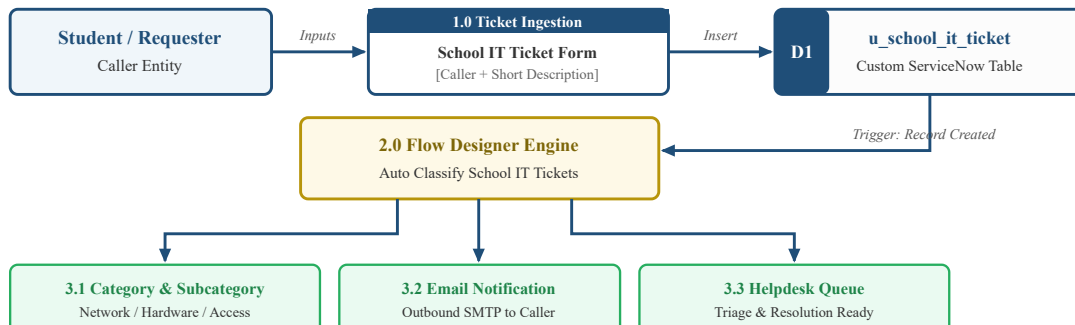


Figure 5: Data flow for the automatic classification process.

Data Flow Diagram (Figure 5) Detailed Breakdown:

- **Ingestion:** Requester inputs Caller identity and Short Description via the Next Experience form.
- **Persistence:** The record is committed to `u_school_it_ticket`, generating an asynchronous database insert event.
- **Evaluation:** Flow Designer triggers on insert, evaluating short description text against keyword condition branches.
- **Action:** Category and Subcategory fields are updated, and an outbound SMTP email acknowledgment is dispatched.

6. TECHNOLOGY STACK

6.1 Platform Architecture

The solution is constructed entirely within the ServiceNow cloud platform, eliminating external infrastructure and middleware dependencies.

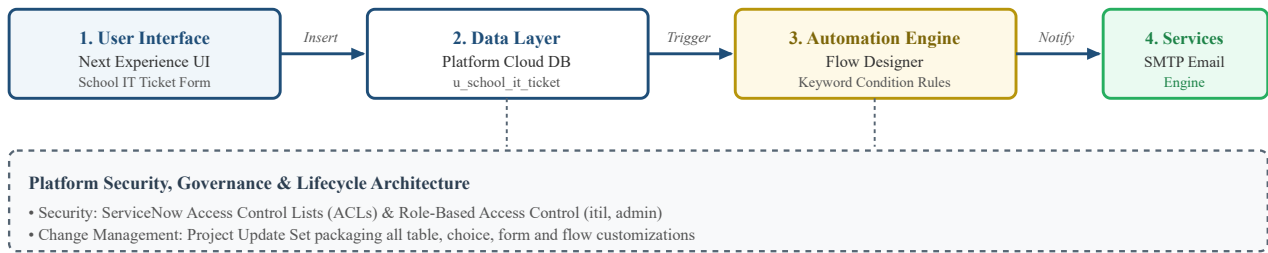


Figure 6: Technical architecture of the ServiceNow-based ticket automation solution.

Technical Architecture (Figure 6) Layer Explanation:

- **User Interface:** ServiceNow Next Experience provides responsive forms and list views.
- **Application Logic:** Flow Designer orchestrates trigger, decision branches, field updates, and notifications.
- **Database:** Native relational database hosts the `u_school_it_ticket` table with dependent choice lists.
- **Security & Change Tracking:** Platform ACLs enforce data integrity, while Update Sets capture all modifications.

7. PROJECT PLANNING & SPRINTS

7.1 Sprint Backlog & Estimation

The project was executed across five focused sprints totaling 37 estimated story points mapped to twelve distinct tasks.

Agile Sprint Delivery Lifecycle

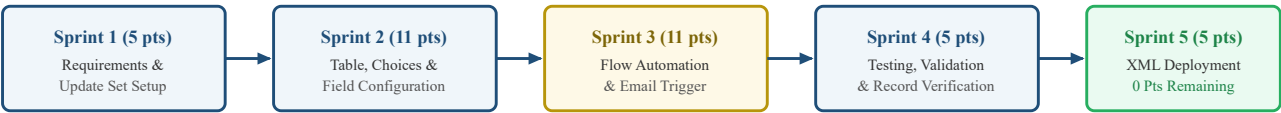


Figure 7: Agile sprint execution lifecycle across five milestones.

7.2 Velocity & Burndown Analysis

Team velocity averaged 7.4 story points per sprint across the 5-sprint lifecycle, successfully delivering all documented scope on schedule.

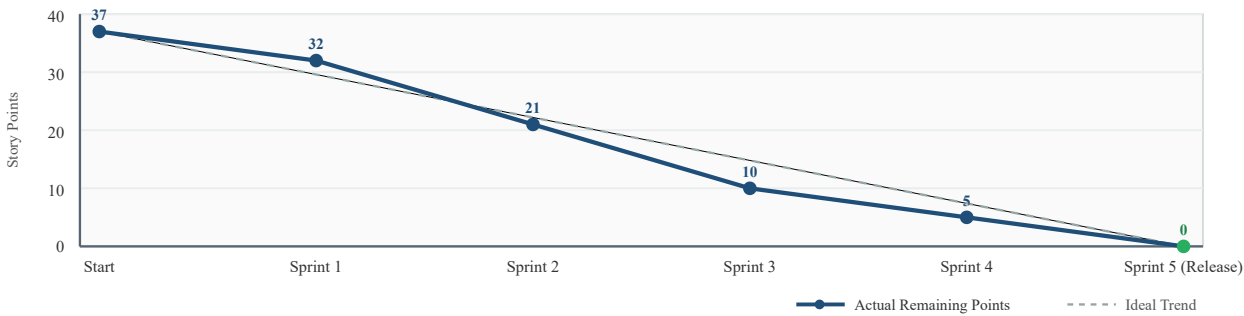


Figure 8: Planned project burndown based on estimated 37 story points.

Burndown Chart (Figure 8) Explanation:

The chart documents constant, predictable progress: 37 initial points burn down through Sprint 1 (32 remaining), Sprint 2 (21 remaining), Sprint 3 (10 remaining), Sprint 4 (5 remaining), and Sprint 5 (0 remaining at final XML release).

8. DEVELOPMENT & CONFIGURATION

8.1 Custom Table and Field Configuration

The `u_school_it_ticket` table was created under the School IT Support application module with standard fields including Number, Caller, Category, Subcategory, Short Description, Description, State, and Assignment Group.

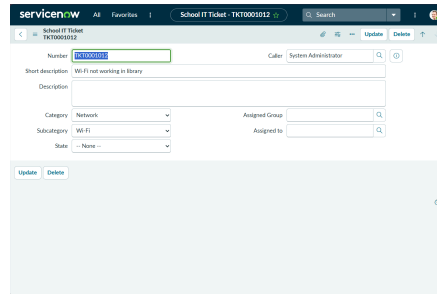


Figure 9: School IT Ticket form showing user interface and entry fields.

8.2 Flow Designer Workflow Implementation

The *Auto Classify School IT Tickets* flow was created and published with a Record Created trigger and conditional logic branches for Wi-Fi, Projector, Password, and Performance issues.

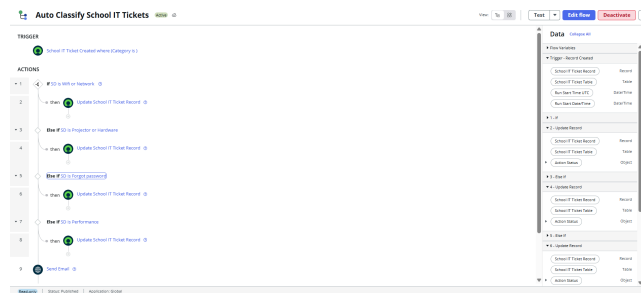
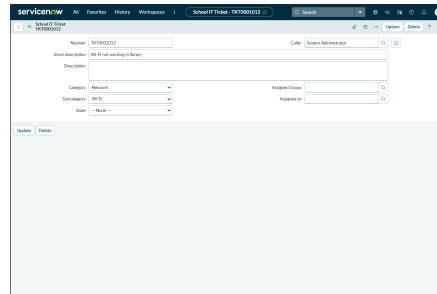


Figure 10: Published Auto Classify School IT Tickets Flow Designer workflow.

9. TESTING & VALIDATION

9.1 User Acceptance Testing

All test cases (TC-001 through TC-005) passed with zero defects recorded. Automatic classification of network and hardware issues was successfully verified.



ServiceNow School IT Ticket: T010000012

Number: T010000012 | Caller: System Administrator

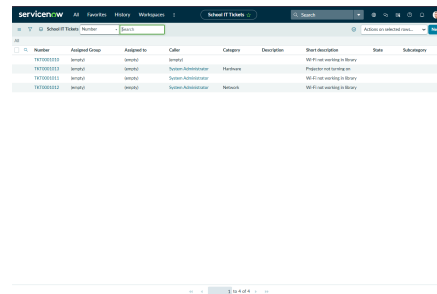
Description: Web site not working in library

Category: Network | Subcategory: Wi-Fi | Assigned To: System Administrator

Status: Open

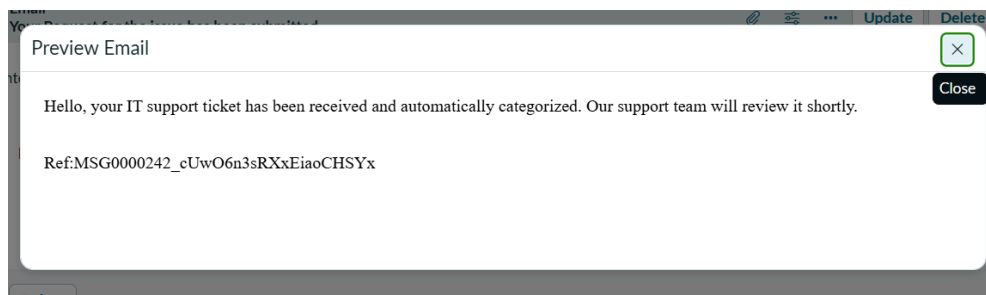
Buttons: Update, Delete

Figure 11: School IT Ticket showing Category = Network and Subcategory = Wi-Fi.



Number	Assigned Group	Assigned To	Caller	Category	Description	Short Description	State	Subcategory
T010000001	Service	Service	Service	Hardware	Problem not in the system	Problem not in the system	Open	
T010000012	Service	Service	System Administrator	Network	Web site not working in library	Web site not working in library	Open	Wi-Fi
T010000013	Service	Service	System Administrator	Network	Web site not working in library	Web site not working in library	Open	Wi-Fi

Figure 12: School IT Tickets list showing created records and classification results.



Preview Email

Hello, your IT support ticket has been received and automatically categorized. Our support team will review it shortly.

Ref:MSG0000242_cUwO6n3sRXxEiaoCHSYx

Buttons: Close, Delete

Figure 13: Preview of the acknowledgment email sent after ticket submission.

10. CONCLUSION & FUTURE SCOPE

10.1 Conclusion

The Auto-Ticket Classification using Flow Designer project demonstrates how no-code platform automation can dramatically simplify service desk interactions. By eliminating requester category guesswork and automating classification, the system reduces technician triage overhead, accelerates incident resolution, and delivers clear communication to students.

10.2 Future Scope

- Expand keyword coverage to include additional academic software and hardware categories.
- Implement automated technician assignment based on Category and technician availability.
- Add resolution and status-change email notifications for callers.
- Develop management dashboards to analyze ticket volume, classification accuracy, and resolution times.
- Explore ServiceNow Predictive Intelligence for AI-assisted classification of edge cases.

11. APPENDIX

Source Code & Configuration Package

The application is implemented as native ServiceNow configurations captured within the Project Update Set. The completed Update Set was exported as an XML package for repository backup and cross-instance deployment.

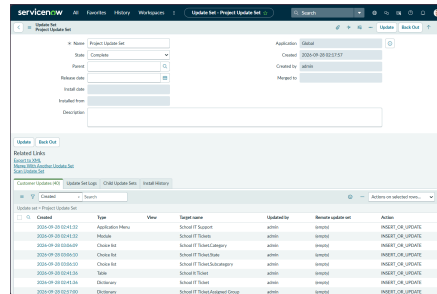


Figure 14: Completed Project Update Set with captured customer updates and the Export to XML option.

Project Links & Repository

GitHub Repository: <https://github.com/Akash717656/Auto-Ticket-Classification-using-Flow-Designer>

Naan Mudhalvan SkillWallet Portal: [Project Review Board](#)

Environment: ServiceNow Developer Instance